THAI POLYMER SUPPLY

PEEK (Polyether ether ketone)



COLORS : Brownish Grey

ADVANTAGES

High mechanical Strength
excellent chemical resistance
very good dimensional stability

APPLICATIONS INCLUDE

Scraper Blades in Haad Exchangers
gas analysis structural body parts
pump wear Rings
Electronics Parts

	Test method	Value
General properties	**ASTM**	
specific Gravity (g/cm^3)	D792	1.31
Water Absorption,24 hrs (%)	D570	0.1
Mechanical properties		
Tensile strength (psi)	D638	16,000
Tensile modulus (psi)	D638	630,000
Tensile Elongation at Break (%)	D638	20
Flexural strength (psi)	D790	25,000
Flexural modulus (psi)	D790	600,000
Compressive strength (psi)	D695	20,000
Compressive modulus (psi)	D695	500,000
Hardness Rockwell	D785	M100
IZOD Notched impact (fi-lb/in)	D256	1
Thermal properties		
Coefficient of thermal expansion(x 10^{-5} in/in/°F)	D696	2.6
Heat deflection temperature (°F/°C)@264 psi	D648	320/160
Melting temperature (°F/°C)	D3418	644/340
Max operating temperature (°F/°C)	-	480/249
Thermal conductivity(BTU-in/ft^2-hr-°F)	C177	1.79
Flammability Rating	UL94	V-0
Electrical properties		
Dielectric strength (V/mil) short time	D149	480
Dielectric constant at 1 MHz	D150	3.3
Dissipation Factor at 1 MHz	D150	0.003
Surface resistivity (0hm/sq) at 50% RH	D257	>10^{13}

THAIPOLYMER

THAIPOLYMER